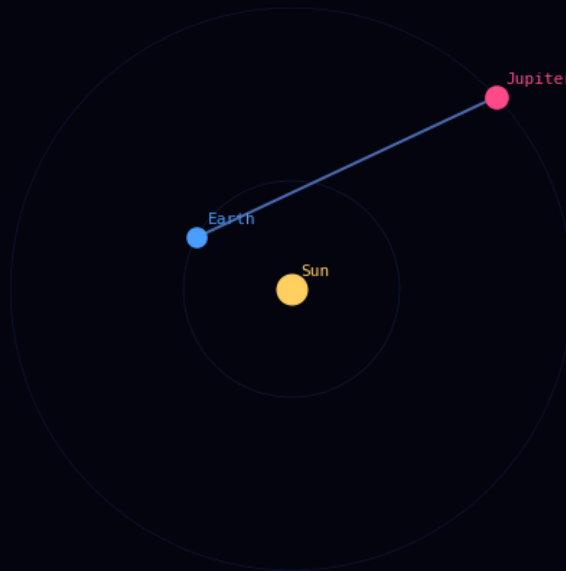


# THE EARTH–JUPITER DANCE

How two planets draw a star-pattern in space



## WHAT IS THIS PATTERN?

Imagine standing in space and drawing a straight line between Earth and Jupiter every single day for ~12 years.

Because Earth orbits FAST (365 days) and Jupiter orbits SLOW (4333 days = 11.86 years), the two planets spin at very different speeds around the Sun.

As time passes, the line between them sweeps through space, tracing a beautiful star-shaped interference pattern.

This PDF builds that pattern step by step — one lap of Earth at a time — so you can watch it emerge.

### KEY NUMBERS

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Earth orbital period:   | 365.25 days (1 year)                                  |
| Jupiter orbital period: | 4332.59 days (11.86 years)                            |
| Ratio:                  | Earth laps Jupiter ~12.0 times during one Jovian year |

# LAP 1 — Earth has orbited 1x, Jupiter moved 30 deg

Days elapsed: 365 of 4333 (8.4% of Jupiter's orbit)

This lap only  
(Earth orbit 1)



Cumulative  
(all 1 lap(s))



## WHAT YOU ARE SEEING — LAP 1

Earth completes its 1st orbit. Jupiter has barely moved — only 30 degrees. A few lines fan out, forming the very first 'spoke' of the pattern.

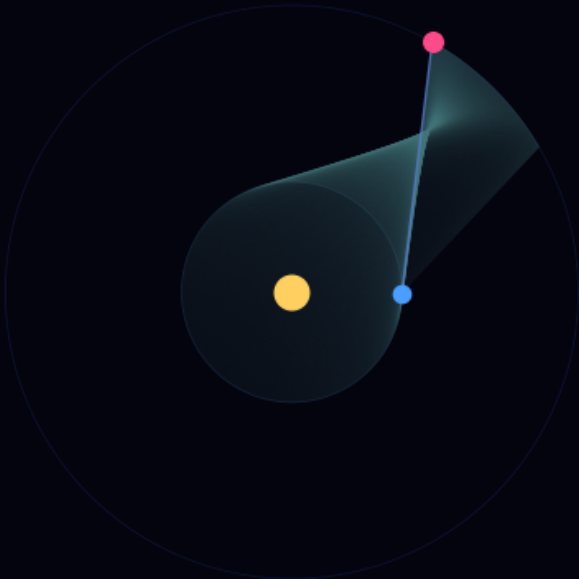
Lines drawn this lap: ~365    Cumulative lines: ~365    Jupiter at: 30.3 deg

- Sun
- Earth
- Jupiter

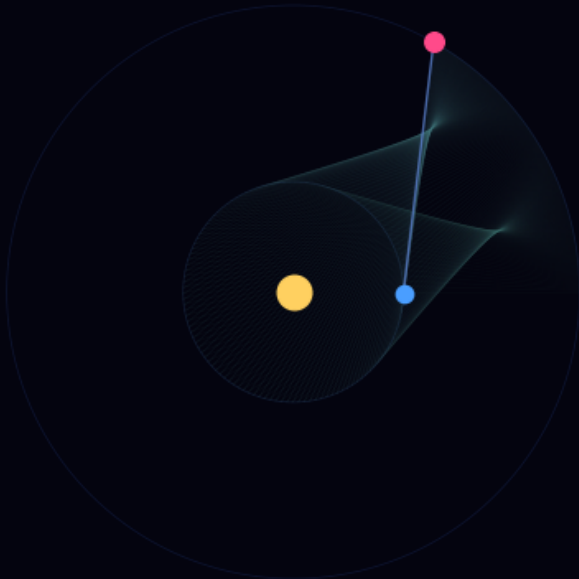
# LAP 2 — Earth has orbited 2x, Jupiter moved 61 deg

Days elapsed: 730 of 4333 (16.9% of Jupiter's orbit)

This lap only  
(Earth orbit 2)



Cumulative  
(all 2 lap(s))



## WHAT YOU ARE SEEING — LAP 2

Earth finishes lap 2. Jupiter has advanced ~60 deg. A second spoke appears, offset from the first. You can already see the pattern wants to be symmetric.

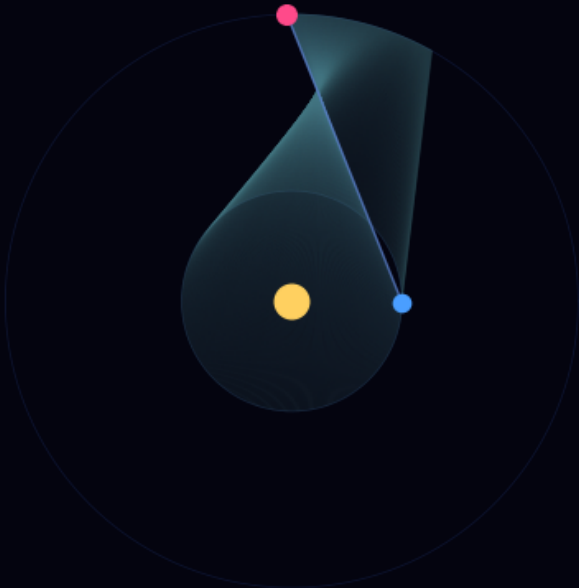
Lines drawn this lap: ~365   Cumulative lines: ~730   Jupiter at: 60.7 deg

- Sun
- Earth
- Jupiter

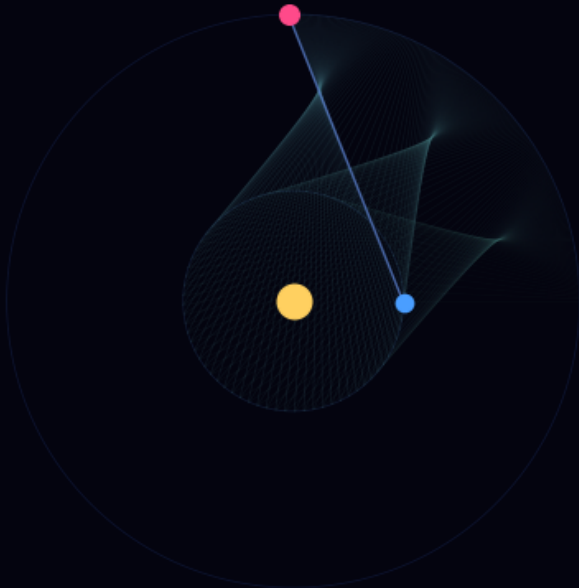
# LAP 3 — Earth has orbited 3x, Jupiter moved 91 deg

Days elapsed: 1095 of 4333 (25.3% of Jupiter's orbit)

This lap only  
(Earth orbit 3)



Cumulative  
(all 3 lap(s))



## WHAT YOU ARE SEEING — LAP 3

Lap 3 done. Jupiter at ~90 deg (one quarter of its orbit). Three spokes are visible and the centre is starting to fill in with a soft glow.

Lines drawn this lap: ~365    Cumulative lines: ~1,095    Jupiter at: 91.0 deg

- Sun
- Earth
- Jupiter

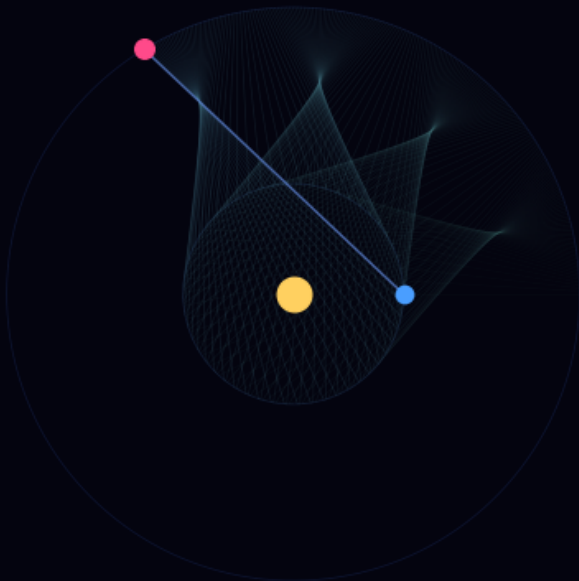
# LAP 4 — Earth has orbited 4x, Jupiter moved 121 deg

Days elapsed: 1461 of 4333 (33.7% of Jupiter's orbit)

This lap only  
(Earth orbit 4)



Cumulative  
(all 4 lap(s))



## WHAT YOU ARE SEEING — LAP 4

Four Earth orbits complete. Jupiter past the 120 deg mark. Four spokes — the star outline begins to feel like a real geometric figure.

Lines drawn this lap: ~365    Cumulative lines: ~1,461    Jupiter at: 121.4 deg

- Sun
- Earth
- Jupiter

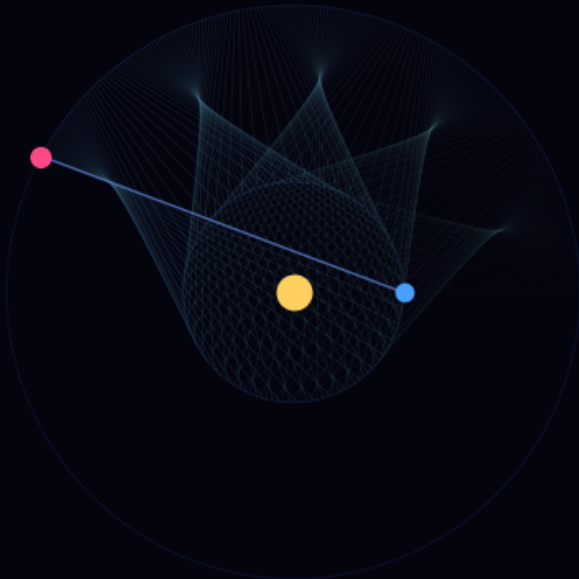
# LAP 5 — Earth has orbited 5x, Jupiter moved 152 deg

Days elapsed: 1826 of 4333 (42.2% of Jupiter's orbit)

This lap only  
(Earth orbit 5)



Cumulative  
(all 5 lap(s))



## WHAT YOU ARE SEEING — LAP 5

Lap 5. Jupiter ~150 deg through its orbit. Five spokes, and the inner star-shape is clearly emerging from the intersection of lines.

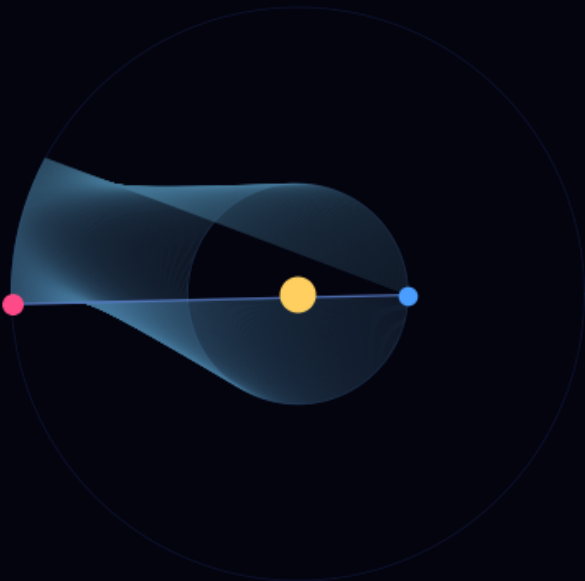
Lines drawn this lap: ~365    Cumulative lines: ~1,826    Jupiter at: 151.7 deg

- Sun
- Earth
- Jupiter

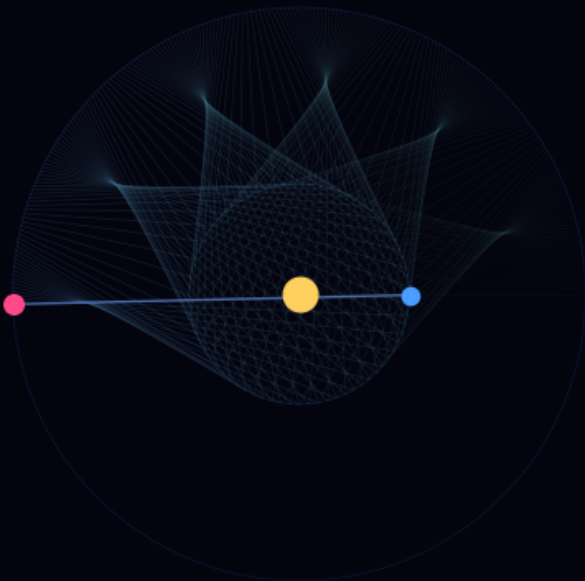
# LAP 6 — Earth has orbited 6x, Jupiter moved 182 deg

Days elapsed: 2191 of 4333 (50.6% of Jupiter's orbit)

This lap only  
(Earth orbit 6)



Cumulative  
(all 6 lap(s))



## WHAT YOU ARE SEEING — LAP 6

Halfway through Jupiter's journey (lap 6, ~180 deg). The pattern has beautiful 6-fold symmetry in progress; the bright core glows strongly.

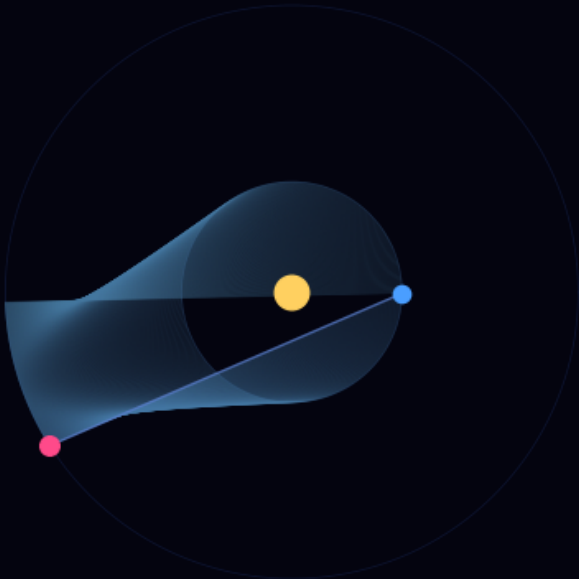
Lines drawn this lap: ~365    Cumulative lines: ~2,191    Jupiter at: 182.1 deg

- Sun
- Earth
- Jupiter

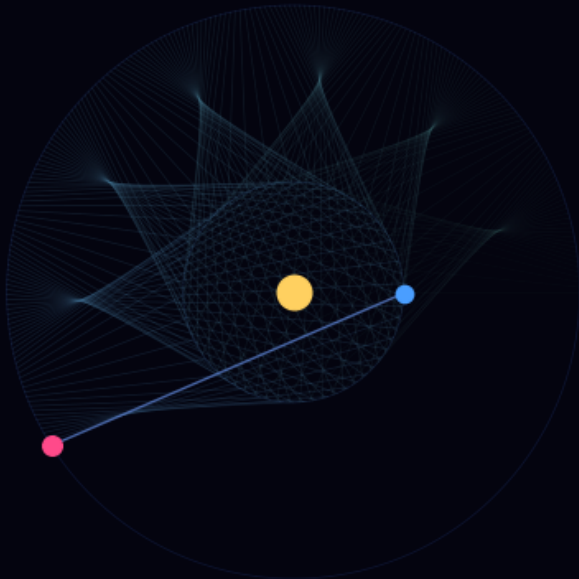
# LAP 7 — Earth has orbited 7x, Jupiter moved 212 deg

Days elapsed: 2556 of 4333 (59.0% of Jupiter's orbit)

This lap only  
(Earth orbit 7)



Cumulative  
(all 7 lap(s))



## WHAT YOU ARE SEEING — LAP 7

Seven Earth laps. Jupiter at 210 deg. The star has 7 prominent arms and the swept arcs on the outer ring are becoming visible.

Lines drawn this lap: ~365    Cumulative lines: ~2,556    Jupiter at: 212.4 deg

- Sun
- Earth
- Jupiter



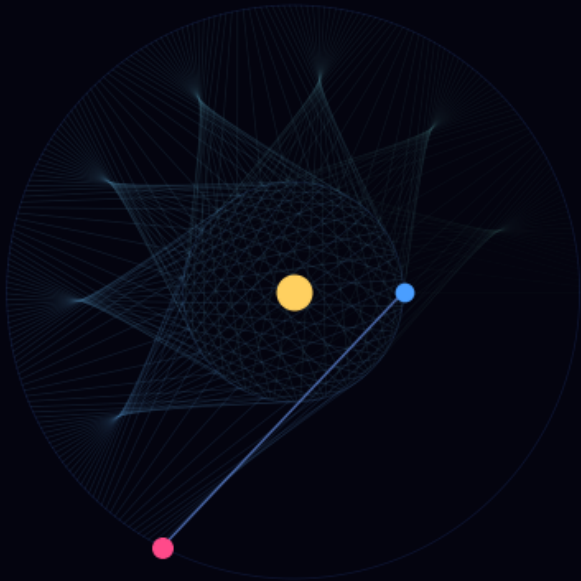
# LAP 8 — Earth has orbited 8x, Jupiter moved 243 deg

Days elapsed: 2922 of 4333 (67.4% of Jupiter's orbit)

This lap only  
(Earth orbit 8)



Cumulative  
(all 8 lap(s))



## WHAT YOU ARE SEEING — LAP 8

Lap 8. Jupiter ~240 deg. Eight spokes; the outer arc segments trace almost a full semicircle — the boundary envelope appears.

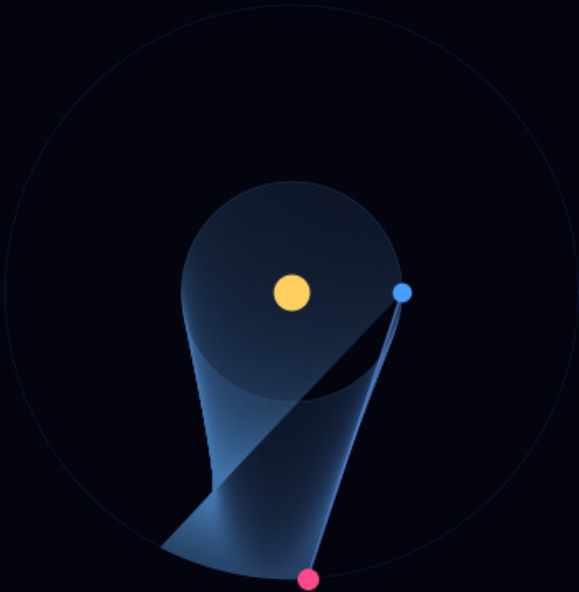
Lines drawn this lap: ~365    Cumulative lines: ~2,922    Jupiter at: 242.8 deg

- Sun
- Earth
- Jupiter

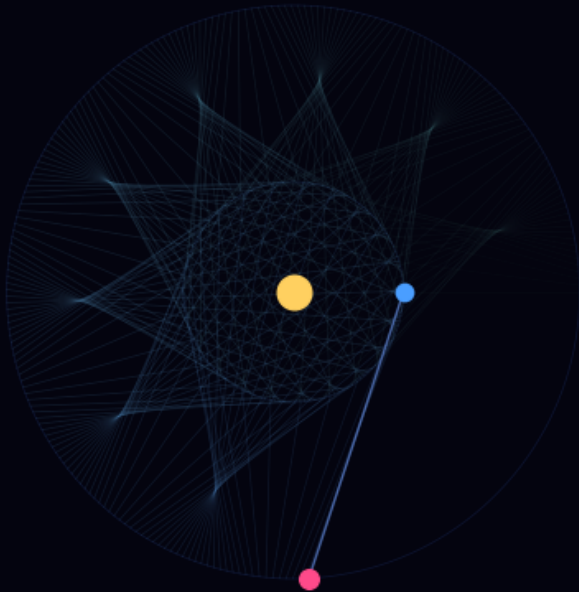
# LAP 9 — Earth has orbited 9x, Jupiter moved 273 deg

Days elapsed: 3287 of 4333 (75.9% of Jupiter's orbit)

This lap only  
(Earth orbit 9)



Cumulative  
(all 9 lap(s))



## WHAT YOU ARE SEEING — LAP 9

Nine Earth orbits. Jupiter at 270 deg (three-quarters done). Nine spokes and the inner star polygon is dense and bright.

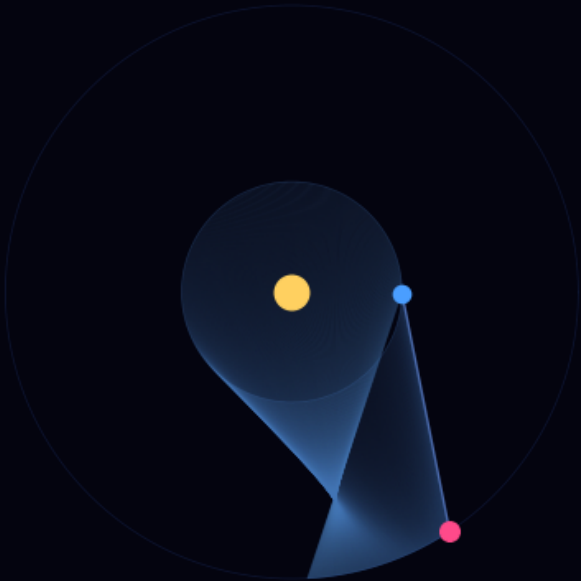
Lines drawn this lap: ~365    Cumulative lines: ~3,287    Jupiter at: 273.1 deg

- Sun
- Earth
- Jupiter

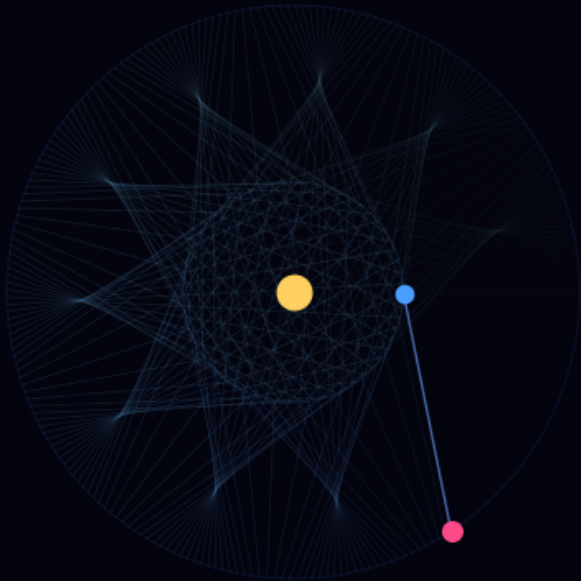
# LAP 10 — Earth has orbited 10x, Jupiter moved 303 deg

Days elapsed: 3652 of 4333 (84.3% of Jupiter's orbit)

This lap only  
(Earth orbit 10)



Cumulative  
(all 10 lap(s))



## WHAT YOU ARE SEEING — LAP 10

Ten laps. Jupiter ~300 deg. The figure is nearly complete; the 10-pointed star is almost fully drawn.

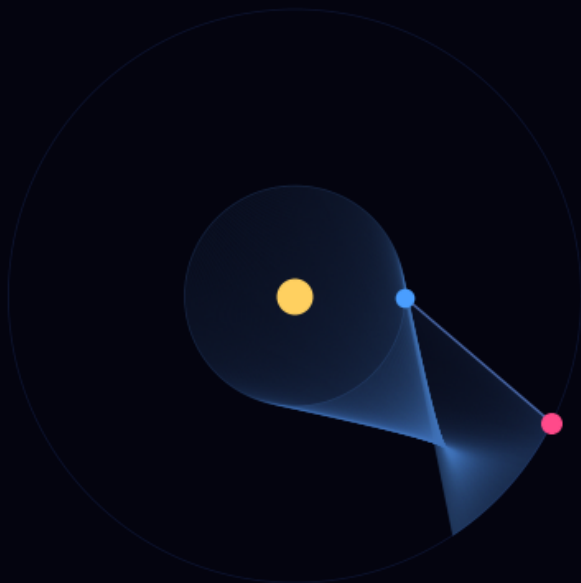
Lines drawn this lap: ~365    Cumulative lines: ~3,652    Jupiter at: 303.5 deg

-  Sun
-  Earth
-  Jupiter

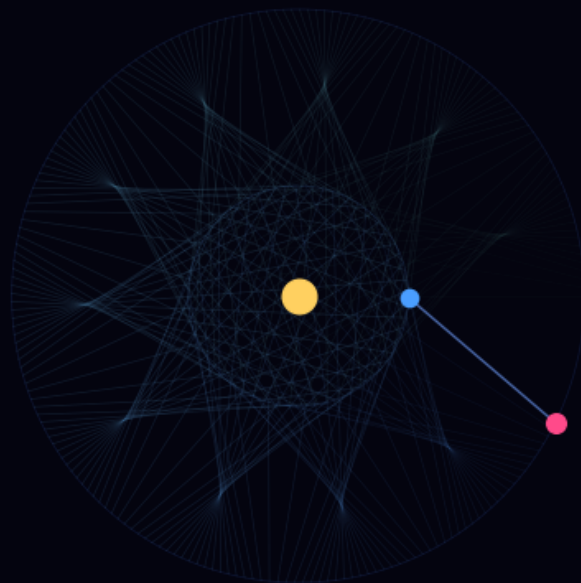
# LAP 11 — Earth has orbited 11x, Jupiter moved 334 deg

Days elapsed: 4017 of 4333 (92.7% of Jupiter's orbit)

This lap only  
(Earth orbit 11)



Cumulative  
(all 11 lap(s))



## WHAT YOU ARE SEEING — LAP 11

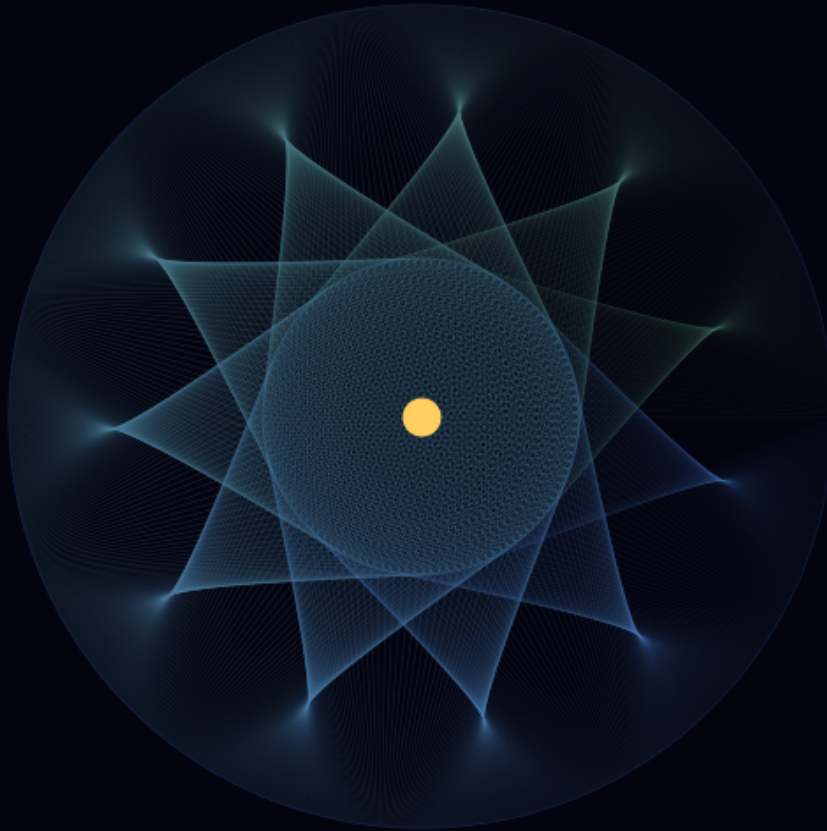
Eleven Earth orbits. Jupiter at ~330 deg — almost home. Eleven spokes, the outer arcs form a nearly closed ring.

Lines drawn this lap: ~365   Cumulative lines: ~4,017   Jupiter at: 333.8 deg

-  Sun
-  Earth
-  Jupiter

# THE COMPLETE PATTERN

All 4332 days — one full Jupiter orbit (~11.86 years)



## THE FINISHED DANCE

After ~12 Earth orbits, Jupiter has completed exactly one of its own.  
The thousands of connecting lines have woven a glowing 11-pointed star.  
The pattern is NOT random — it is pure geometry arising from the ratio of the two orbital periods:  $4332.59 / 365.25 \approx 11.86$ .  
Each "petal" of the star corresponds to one Earth orbit.

- Sun
- Earth
- Jupiter

# THE MATHEMATICS

Why does the pattern have ~11 arms?

## 1. SYNODIC PERIOD

The synodic period is the time between two moments when Earth and Jupiter are in the same relative alignment (both on the same side of the Sun).

$$\begin{aligned}1/P_{\text{syn}} &= 1/P_{\text{Earth}} - 1/P_{\text{Jupiter}} \\1/P_{\text{syn}} &= 1/365.25 - 1/4332.59 \\P_{\text{syn}} &= 398.9 \text{ days} \quad (\sim 1.09 \text{ years})\end{aligned}$$

This means Earth "laps" Jupiter every 399 days.

## 2. NUMBER OF LAPS IN ONE JUPITER ORBIT

In the time Jupiter completes one full orbit (4332.59 days), Earth goes around  $4332.59 / 365.25 = 11.86$  times.

The fractional part 0.86 means Earth is NOT in a perfect integer resonance with Jupiter. If it were exactly 12 laps, the star would close perfectly. Instead it drifts slightly each time.

## 3. WHY ~11 STAR-ARMS?

Each Earth orbit, the "current position" of Jupiter advances by  $360 / 11.86 \approx 30.3$  degrees.

So successive lap-start lines are offset by ~30 deg.  
11 laps later, Jupiter has moved  $11 \times 30.3 \approx 333$  deg,  
and the 12th lap closes the figure — giving 11 distinct arms.

## 4. THE ENVELOPE CURVE

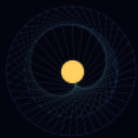
The outer curved boundary (the "petals") is an epicycloid.  
It is traced by a point on a circle of radius  $R_E$  rolling outside a circle of radius  $R_J - R_E$ .

The inner bright star is formed by the INTERSECTIONS of the lines.

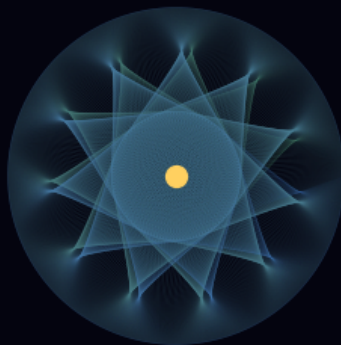
# OTHER PLANETARY DANCES

Every pair of planets draws its own unique pattern

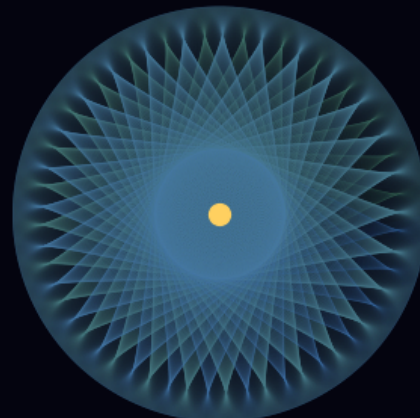
Earth–Venus



Earth–Jupiter



Earth–Saturn



## SUMMARY — WHAT YOU HAVE LEARNED

1. Earth and Jupiter both orbit the Sun, but at very different speeds.
2. A line drawn between them every day sweeps through space over time.
3. After 11.86 years (one Jupiter orbit), the lines form a star-shaped interference pattern — the "Earth-Jupiter Dance".
4. The pattern arises purely from the **RATIO** of orbital periods (~11.86).
5. Every pair of planets draws its own unique locus — Venus makes a 5-petalled rose (almost perfect 8:13 resonance), Saturn draws a denser, more complex web.
6. These patterns are an example of **LISSAJOUS / SPIROGRAPH** geometry that appears all across nature and physics.